

CERTAV: MANIFOLD-AWARE CERTIFIED ROBUSTNESS FOR AUDIO-VISUAL DEEPPFAKE DETECTION

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ABSTRACT

Audio-visual deepfake detectors rely on frozen foundation representations, yet their robustness is typically evaluated only empirically. We introduce CertAV, a certification framework that applies randomized smoothing in the joint feature space of frozen DINOv2 and Whisper encoders. On FakeAVCeleb, CertAV reaches 92.5% clean accuracy and 88.0% certified accuracy at ℓ_2 radius 1.0, with less than 1% abstention. We prove that, under a subspace-invariance property of the detector head, the smoothed classifier’s certified radius depends only on the d -dimensional on-manifold component of the noise ($d \approx 80 \ll D = 768$), implying a robustness amplification factor of $\sqrt{D/d}$. Validating this across five encoder families, we confirm that a lower intrinsic-dimension ratio d/D yields larger certified radii. Exploiting this geometry, we propose manifold-aware anisotropic smoothing that, at equalized noise budget, achieves a $3.4\times$ larger on-manifold certified radius (7.63 vs. 2.22) with zero abstention. The same geometry explains an ablation finding: a no-noise baseline certifies within 1.9% of the noise-trained model, indicating that certifiability is set primarily by representation geometry rather than by the training procedure.

Index Terms— audio-visual deepfake detection, certified robustness, randomized smoothing, anisotropic smoothing, media forensics

1. INTRODUCTION

Audio-visual deepfake detection has moved from single-stream artifact recognition toward cross-modal reasoning: whether a face, voice, expression, and temporal speech pattern belong to the same event. Modern detectors increasingly build on frozen self-supervised encoders, and while they are accurate under clean evaluation, they can remain vulnerable to small adversarial feature shifts. For forensic use, a calibrated detector is useful; a detector with a per-sample robustness certificate is stronger.

Randomized smoothing converts a base classifier into a smoothed classifier whose prediction is provably unchanged within an ℓ_2 ball around its input [1]. Its common use is image-space classification, where the radius is measured in pixels. Audio-visual deepfake detection is less straightforward: videos are long, audio is continuous,

and detectors typically consume frozen DINOv2 [2] and Whisper [3] features rather than raw input. CertAV therefore certifies the detector in this joint feature space. This is a deliberately scoped guarantee: it does not certify arbitrary waveform or pixel perturbations, but gives an exact robustness statement for the representation actually consumed by the detector.

The central observation is that feature-space smoothing gives large certified radii with little abstention, and that this is not primarily an effect of noise-augmented training. Instead, the frozen multimodal representation is low-dimensional relative to its ambient feature space, so isotropic Gaussian noise can move a sample substantially without crossing the decision boundary. We make this precise: under a subspace-invariance property of the detector head, the certified radius is governed only by the on-manifold noise, and the gap between an adversary’s ambient budget and its on-manifold effect scales as $\sqrt{D/d}$. This explains the observed radii, predicts which encoders certify best, and motivates a smoothing distribution aligned to the manifold.

This paper makes four contributions. (i) We define CertAV, a feature-space randomized-smoothing pipeline for audio-visual deepfake detection under a joint ℓ_2 threat model. (ii) We prove a *Certifiability Scaling* theorem and a *Robustness Amplification* corollary that explain why frozen multimodal features certify well and give a principled encoder-selection criterion. (iii) We validate the theory with an *encoder-scaling study* across five visual/audio encoder families, confirming that lower d/D yields larger certified radii. (iv) We propose *manifold-aware anisotropic smoothing*, which at equalized noise budget achieves a $3.4\times$ larger on-manifold certified radius with zero abstention.

2. RELATED WORK

Audio-visual deepfake detection. Audio-visual detectors exploit inconsistencies invisible in either modality alone, from affective disagreement [4] and audio-visual dissonance [5] to self-supervised anomaly detection [6], joint learning [7], and local temporal inconsistency modeling [8]. Datasets such as FakeAVCeleb [9] and LAV-DF [10] provide multimodal settings. These works improve accuracy and localization, but their robustness claims are largely empirical.

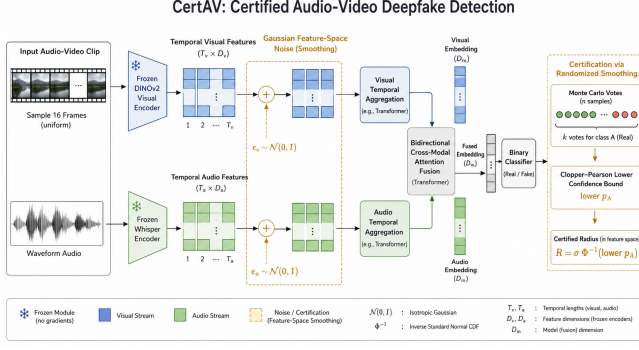


Fig. 1. CertAV pipeline. Frozen DINOv2 and Whisper encoders produce temporal visual and audio features, which are aggregated, fused by bidirectional cross-modal attention, and certified by randomized smoothing in joint feature space.

Certified robustness. Randomized smoothing certifies an ℓ_2 ball by estimating the probability that a Gaussian-perturbed base classifier predicts the top class [1, 11, 12]. Yang et al. [13] generalize this to arbitrary noise distributions, showing that non-isotropic smoothing yields ellipsoidal certified regions, and ANCER [14] optimizes per-sample anisotropic noise to maximize certified volume. MMCert [15] certifies multimodal models by randomized ablation of raw inputs. CertAV differs by certifying the frozen *feature* space consumed by the detector, by giving a theoretical account of why this space certifies well, and by using a simple global PCA-aligned anisotropic distribution justified by that account rather than per-sample volume optimization.

3. METHOD

3.1. Feature representation and detector

Given an audio-video clip $\mathbf{x}$, CertAV extracts visual and audio feature sequences $Z_v = E_v(\mathbf{x}_v)$ and $Z_a = E_a(\mathbf{x}_a)$, where E_v is a frozen DINOv2-Small encoder (16 frames, 384-d embeddings) and E_a is a frozen Whisper-tiny encoder (16 kHz audio up to 10 s, 384-d temporal embeddings). The base detector f_θ projects each modality to a 256-d temporal sequence (a transformer aggregator for video, temporal pooling for audio), applies a two-layer bidirectional cross-modal attention module, and averages segment logits before a binary real/fake decision. Features are cached and used as the input domain for training, attacks, and certification.

3.2. Threat model and certification

Let $\mathbf{z} = (Z_v, Z_a) \in \mathbb{R}^D$ ($D = 768$) denote the concatenated frozen representation. CertAV certifies robustness to joint feature perturbations δ with $\|\delta\|_2 \leq r$. This is the correct abstraction for frozen-encoder pipelines, where the detector never observes raw input: certifying the giant frozen encoders end-to-end is computationally intractable and yields vanishingly small radii, whereas the feature-space certificate is exact for any adversary acting on the representation the detector consumes. To connect the two spaces we additionally measure an empirical feature-to-input Lipschitz constant L from input-space attacks and compose it with the feature certificate; on FakeAVCeleb $L \approx 50$, mapping the on-manifold certificate to an approximate input-space radius of ≈ 0.12 at $\approx 90.7\%$ composed

certified accuracy. The smoothed classifier is

$$g(\mathbf{z}) = \arg \max_c \Pr_{\epsilon \sim \mathcal{N}(0, \sigma^2 I)} (f_\theta(\mathbf{z} + \epsilon) = c). \quad (1)$$

During training, Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ is injected into the features and the model is fit with binary cross-entropy. Certification follows the two-stage Monte Carlo procedure of [1]: $n_0 = 100$ samples select the candidate class, $n = 1000$ samples estimate its probability, and a Clopper-Pearson lower bound $\underline{p}_A$ ($\alpha = 0.001$) gives the certified radius $R_{\text{cert}} = \sigma \Phi^{-1}(\underline{p}_A)$, abstaining if $\underline{p}_A \leq 0.5$.

3.3. Manifold-aware anisotropic smoothing

Let $\mathcal{S}$ be the d -dimensional subspace spanned by the top principal components of the training features (the 90%-variance threshold gives $d \approx 80$). Anisotropic smoothing replaces isotropic $\mathcal{N}(0, \sigma^2 I)$ by $\mathcal{N}(0, \Sigma)$ with Σ diagonal in the PCA basis. All strategies preserve the *total* noise budget, $\text{trace}(\Sigma) = D\sigma^2$, so improvements come from *redistributing* noise, not adding it. We report two radii: the worst-case ℓ_2 radius $R_{\ell_2} = \Phi^{-1}(\underline{p}_A) \sqrt{\lambda_{\min}(\Sigma)}$ (largest inscribed sphere), and the on-manifold radius $R_{\mathcal{S}} = \Phi^{-1}(\underline{p}_A) \sqrt{\bar{\lambda}_{\mathcal{S}}}$, where $\bar{\lambda}_{\mathcal{S}}$ is the mean variance over the top- d directions in which the decision boundary lies.

4. THEORETICAL ANALYSIS

Let $P_{\mathcal{S}}$ project onto $\mathcal{S}$ and $P_{\perp} = I - P_{\mathcal{S}}$ onto its complement.

Assumption 1 (Subspace invariance). *For all $\mathbf{z} \in \mathcal{S}$ and all $\delta \in \mathbb{R}^D$, $f(\mathbf{z} + \delta) = f(\mathbf{z} + P_{\mathcal{S}}\delta)$: only the on-manifold component of a perturbation affects the classifier.*

This is an idealization, justified because f is trained on features drawn from $\mathcal{S}$ and receives no gradient from off-manifold directions; we validate it empirically below.

Theorem 1 (Certifiability scaling). *Under Assumption 1, for any $\mathbf{z} \in \mathcal{S}$ the top-class probability p_A of the smoothed classifier with $\epsilon \sim \mathcal{N}(0, \sigma^2 I_D)$ depends on ϵ only through its projection $\epsilon_{\mathcal{S}} = P_{\mathcal{S}}\epsilon \sim \mathcal{N}(0, \sigma^2 I_d)$, and the certified radius is $R_{\text{cert}} = \sigma \Phi^{-1}(p_A)$.*

Proof sketch. Decompose $\epsilon = \epsilon_{\mathcal{S}} + \epsilon_{\perp}$. Because $P_{\mathcal{S}}$ and $P_{\perp}$ are complementary orthogonal projections, $\epsilon_{\mathcal{S}}$ and $\epsilon_{\perp}$ are independent Gaussians. By Assumption 1, $f(\mathbf{z} + \epsilon) = f(\mathbf{z} + \epsilon_{\mathcal{S}})$ for every $\epsilon_{\perp}$, so p_A is determined by the d -dimensional variable $\epsilon_{\mathcal{S}}$; the radius then follows from [1]. $\square$

Corollary 1 (Robustness amplification). *For an adversary with budget $\|\delta\|_2 \leq r$ and direction uniform on $\mathbb{S}^{D-1}$, $\mathbb{E} \|P_{\mathcal{S}}\delta\|_2^2 = (d/D) r^2$, so achieving on-manifold displacement m requires $r \geq m \sqrt{D/d}$. The amplification factor is $A = \sqrt{D/d}$; among encoders with the same D , those with smaller d/D are more certifiable.*

The proof computes $\mathbb{E}[u_i^2] = 1/D$ for a uniform direction u and sums over the d on-manifold coordinates. Theorem 1 explains why a no-noise baseline certifies almost as well as the noise-trained model: the frozen geometry, not the training procedure, fixes the effective on-manifold noise. Corollary 1 converts this into an encoder-selection rule and motivates anisotropic smoothing, since noise placed off-manifold is wasted.

Table 1. Joint feature-space certification on FakeAVCeleb over five seeds. Accuracy, abstention, and certified accuracy ($C@r$) are percentages.

σ	Acc.	Abs.	Mean $\bar{R}$	$C@0.25$	$C@0.50$	$C@1.00 / 1.50$
0.12	91.2	0.3	0.288	88.8	0.0	0.0 / 0.0
0.25	91.2	0.4	0.588	89.4	86.6	0.0 / 0.0
0.50	91.5	0.4	1.165	90.5	89.2	85.4 / 0.0
1.00	92.5	0.8	2.215	91.8	90.7	88.0 / 84.9

Table 2. Encoder-scaling study (no-noise baseline, $\sigma = 1.0$). d is the 90%-variance PCA dimension; $\mathcal{A} = \sqrt{D/d}$. Lower d/D tracks larger $\bar{R}$.

Encoder pair	D	d	d/D	$\mathcal{A}$	$\bar{R}$
DINOv2-S + WavLM	1152	80	0.069	3.79	2.207
DINOv2-S + HuBERT	1152	79	0.069	3.82	2.173
DINOv2-B + Whisper-base	1280	119	0.093	3.28	2.132
DINOv2-S + Whisper-tiny	768	80	0.104	3.10	2.190
CLIP-B + Whisper-tiny	1152	126	0.109	3.02	1.948

5. EXPERIMENTS

Setup. FakeAVCeleb [9] is split into 3850 train, 825 validation, and 825 test clips (1:10 real/fake). Main results average five seeds (42, 69, 420, 2026, 2804); models use AdamW (lr 5×10^{-4} , weight decay 0.01), batch size 8, cosine decay, and early stopping. Certification uses $n_0 = 100$, $n = 1000$, $\alpha = 0.001$.

Isotropic certification. Table 1 shows that larger smoothing noise increases the certified radius without lowering clean accuracy, and abstention stays below 1%. At $\sigma = 1.0$, CertAV certifies 84.9% of the test set at radius 1.5 — atypical for a detector merely memorizing clean artifacts, and consistent with Theorem 1. A no-noise baseline reaches mean radius 2.173 (within 1.9% of the noise-trained 2.215), while feature-space PGD adversarial training raises the radius to 2.463 but collapses validation AUC from 0.941 to 0.730 (EER 0.139 $\rightarrow$ 0.335): it trivially stabilizes predictions at the cost of discrimination. Certifiability is therefore a property of the representation geometry, not of the training recipe.

Encoder scaling. Table 2 evaluates five encoder pairs (using WavLM [16], HuBERT [17], and CLIP [18] variants). The trend predicted by Corollary 1 holds at the extremes: the lowest d/D pairs (DINOv2-S + WavLM/HuBERT, $d/D \approx 0.069$) attain the largest radii ($\bar{R} \approx 2.2$), while CLIP-B + Whisper-tiny ($d/D = 0.109$) attains the smallest (1.948) despite the highest clean accuracy (94.3%). The ordering is not strictly monotone in the mid-range, so d/D is the dominant but not sole predictor; clean accuracy and certified radius can trade off.

Anisotropic smoothing (headline). Table 3 reports three PCA-aligned strategies at *equalized* budget. Concentrating noise in the decision-relevant subspace (Strategy 2, subspace projection) yields an on-manifold radius of 7.63, a $3.4\times$ lift over isotropic (2.22), and attains 0% abstention — every correctly classified sample is certified. The near-zero R_{ℓ_2} is by design (Fig. 2): the certificate is an ellipsoid with long axes in the 80 on-manifold directions and tiny axes in the 688 off-manifold directions where the classifier is invariant. A diagnostic confirms the premise: PGD attack directions place only $\cos^2 \theta \approx 0.18$ of their energy on-manifold, so 82% of any ambient attack is wasted, exactly the regime Theorem 1 predicts.

Transfer and degradations. On LAV-DF [10], CertAV transfers at 74.6% accuracy and 61.4% certified accuracy at radius 1.0, indicating useful but imperfect cross-dataset generalization. Common media degradations (social-media re-encoding, H.264 CRF 28, JPEG-

Table 3. Manifold-aware anisotropic smoothing at equalized budget trace(Σ) = $D\sigma^2$ ($\sigma = 1.0$). For anisotropic rows, $C_S@1.0$ is certified accuracy w.r.t. the *on-manifold* radius R_S (not the worst-case R_{ℓ_2}); for the isotropic row the two radii coincide. Strategy 2 gives the largest on-manifold radius and zero abstention.

Strategy	Acc.	Abs.	R_{ℓ_2}	R_S	$C_S@1.0$
Isotropic	90.9	0.8	2.215	2.215	88.0
S1: eigenvalue-prop.	90.9	0.7	0.000	6.272	89.5
S2: subspace proj.	90.9	0.0	0.002	7.632	90.9
S3: inverse-eigenvalue	90.9	0.7	0.014	6.586	89.0

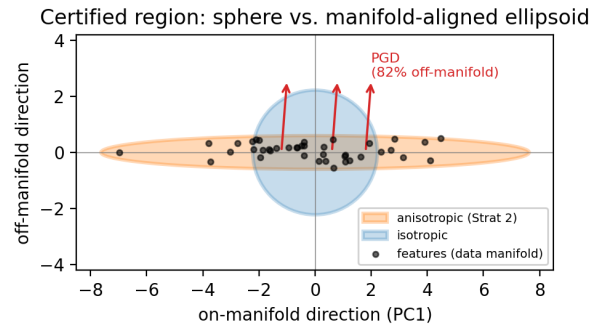


Fig. 2. Isotropic smoothing certifies a sphere (equal in all directions); manifold-aware smoothing certifies an ellipsoid that extends far along the data manifold and little off it. PGD attacks are 82% off-manifold ($\cos^2 \theta \approx 0.18$), where the classifier is provably invariant.

75) preserve $> 90\%$ certified accuracy at radius 0.5. Unimodal certificates can have larger numerical radii but protect only one stream.

6. CONCLUSION

CertAV makes certified robustness practical for audio-visual deep-fake detection by placing the guarantee at the level of frozen multi-modal features. Its central lesson is geometric: the intrinsic dimension of the representation, not the training procedure, governs certifiability (Theorem 1), which both predicts encoder behavior and motivates manifold-aware anisotropic smoothing for a $3.4\times$ on-manifold gain. The evaluation logic is released as *av-robustbench*, a reproducible scaffold for attacks, certificates, degradations, and transfer reporting. The main limitations are the feature-space scope of the certificate (our empirical Lipschitz composition is a bridge, not a worst-case input proof), the approximate rather than exact subspace invariance underlying Theorem 1 ($\cos^2 \theta = 0.18 \neq 0$), single-dataset evaluation, and the cost of Monte Carlo certification.

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